

Hunting Primates and Managing Forests: The Case of Iban Forest Farmers in Indonesian Borneo

Reed L. Wadley,¹ Carol J. Pierce Colfer,² and Ian G. Hood³

Hunting by Iban forest farmers in West Kalimantan, Indonesia, is an important part of their subsistence economy, and as such became a focus of study as part of a conservation project in the Danau Sentarum Wildlife Reserve. In this paper, we examine Iban hunting of nonhuman primates with comparison to other large mammals. We analyze rates of encounter and capture, comparing encounters, hunting trips, and animal numbers. Information on habitats hunted shows the importance of secondary and old growth forest. Also examined are Iban attitudes, game preferences, and taboos. The significance of these findings is discussed with regard to the threats to wildlife from increases in the use of shotguns, human population, and habitat destruction, showing that conservation may be aided by promoting or enhancing certain aspects of the traditional Iban agroforestry system.

KEY WORDS: wildlife conservation; primatology; hunting; indigenous management; Indonesia.

INTRODUCTION

Hunting in tropical forests by farmers has increasingly become a focus of study by researchers interested in conservation issues (e.g., Caldecott, 1988; Stearman, 1992; Grenand, 1993; Lahm, 1993a,b; Redford, 1993; Takeda and Sato, 1993; Vickers, 1993; Jorgenson, 1995; Lehman *et al.*, 1995). The interaction between sedentary or semi-sedentary peoples and tropical forests is important for a number of reasons: First, forest farmers⁴ have a significant impact on forest resources through their land use prac-

¹Department of Anthropology, Arizona State University, Tempe, Arizona 85287.

²Center for International Forestry Research, Jl. Gunung Batu 5, Bogor, Indonesia.

³Mfangano Island, Musiara Camps, P.O. Box 48217, Nairobi, Kenya.

⁴Forest farming refers to the practice of swidden cultivation with a long fallow, in which a plot of forest is felled, cultivated for a year or two, and then allowed to return to forest growth before being farmed again (see e.g., Conklin, 1957; Freeman, 1970; Dove, 1981).

tices; second, they have an intimate and detailed knowledge of the forest flora and fauna that are deemed precious by governments and conservationists; and third, they can play a vital role in forest and wildlife conservation management.

In this paper, we examine hunting among a community of Iban forest farmers in West Kalimantan, Indonesia, with a special focus on the hunting of nonhuman primates.⁵ Primates occupy a unique place in the attention of people around the world. There is their physical resemblance to humans, which features prominently in the popular media and imagination; their close genetic relationship to humans, which is recognized by the scientific community; and their social complexity. There is also what Marsh and Mittermeier (1987, p. xii) call the "flagship" nature of primate species, that given their different ecologies they are good indicators of the condition of tropical forests. It is primarily for this last reason that we focus here on the hunting of primates with comparison to other large mammals.

We compare primates encountered, shot at, and captured to other large mammals, particularly highly valued game species. In addition, the habitats within which encounters occur are examined. Iban attitudes, cultural preferences, and taboos regarding primates are also discussed. The major human factors affecting wildlife in the area are considered, and finally suggestions on the conservation and management of wildlife in the area made with an emphasis on traditional Iban agroforestry practices.

STUDY SITE

Danau Sentarum Wildlife Reserve (DSWR) is located in a remote area of the Indonesian province of West Kalimantan. It is situated close to the Malaysian border of Sarawak and approximately 700 km inland from the provincial capital, Pontianak (Fig. 1). DSWR is an area of interconnected seasonal lakes and seasonally flooded tropical forests. The water catchment consists of lowland tropical forest in the hills and flooded forest in the low lying areas. The former is characterized by a patchwork of various forest stages as a result of both commercial logging and swidden cultivation.

The altitude within the reserve is approximately 30–35 meters above sea level, while the surrounding hills rise to 760 meters. Daytime temperatures are consistently 26–30 degrees Celsius, and annual rainfall ranges between 3000 and 4000 mm. The driest months are usually July, August, and September.

Reserve status was granted in 1982 with additional legislation in 1985 (Giesen, 1987). The main part of the reserve comprises some 800 km², but

⁵Hereafter "primates" will refer to "nonhuman primates" for convenience.

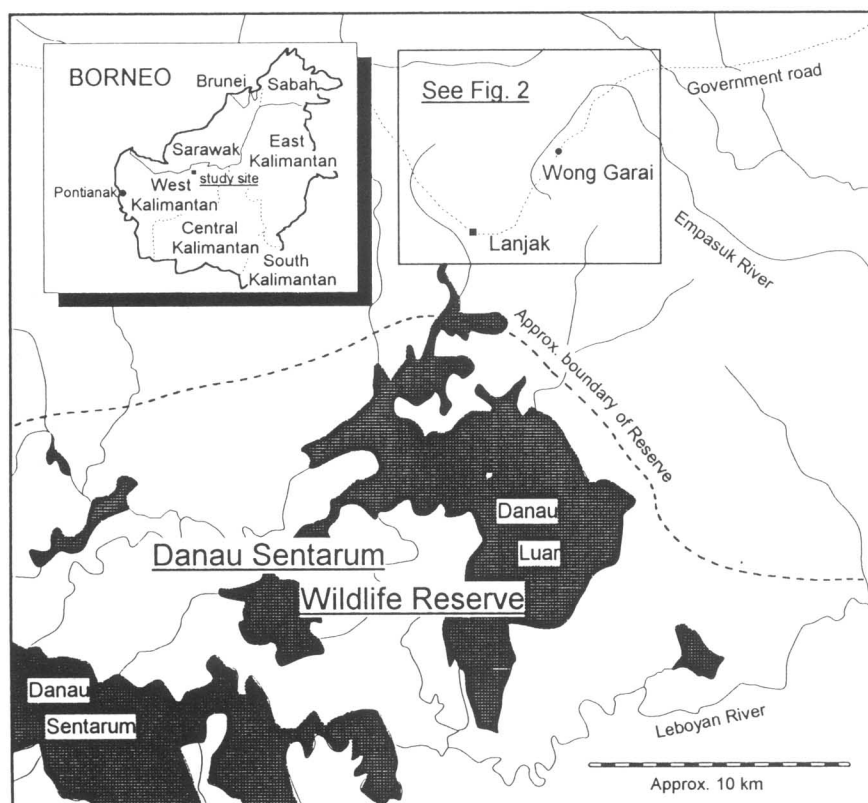


Fig. 1. The Northeast periphery of Danau Sentarum wildlife reserve.

official approval of the boundaries has yet to be attained. There is an effort to expand the reserve boundaries to include an additional 450 km² along the lower Leboyan River (see Deschamps, 1995). As the boundaries of the reserve remain unclear, our study area (located in the hills northeast of the reserve core) may eventually be considered a transition zone, a buffer zone, or even part of the reserve core.

The reserve core consists of a network of seasonally flooded lakes in which some 3000 Muslim Melayu fisherfolk reside. The reserve periphery and surrounding hills are inhabited largely by Christian Dayaks,⁶ the majority of whom are Iban. Population density in the area is approximately 3–4 persons per km².

⁶“Dayak” refers to the indigenous, non-Muslim inhabitants of Borneo; Dayaks who have converted to Islam generally become reclassified as Melayu (see King, 1978).

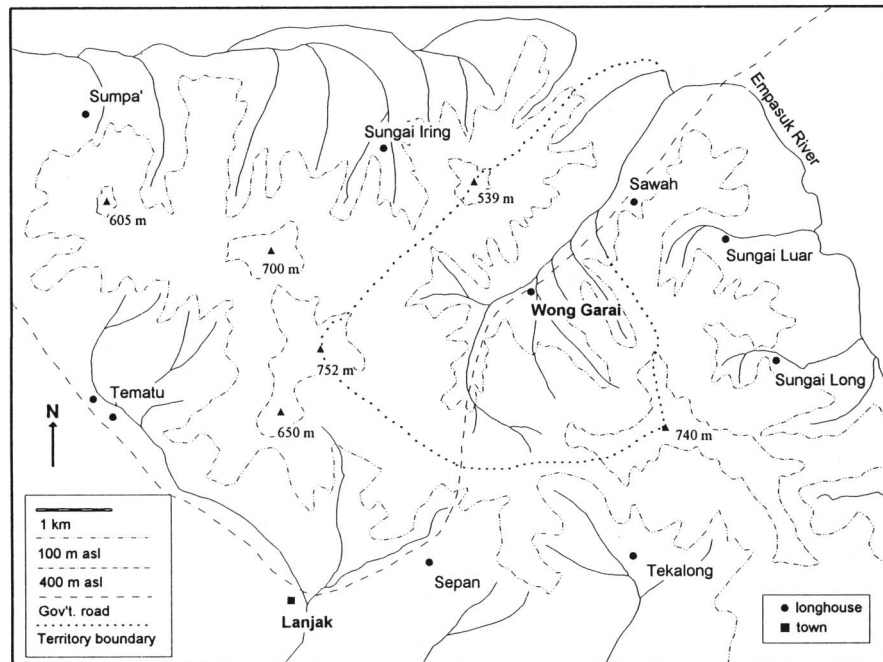


Fig. 2. Wong Garai and its vicinity.

Study focused on the Iban longhouse community of Wong Garai (a pseudonym) which is located about 6 km from Lanjak, the district administrative center and market town (Fig. 2). The longhouse is a 14-household community containing around 98 residents during the period of fieldwork. Average household size is six people. The surrounding territory claimed by the longhouse encompasses approximately 24 km² and is a patchwork of forest succession, agricultural plots, and specially preserved forest.

The Iban practice a complex swidden cultivation system of which hunting is an integral part. Generally characterized in the past as pioneering rice swidden cultivators (see e.g., Freeman, 1970), the Iban today practice more established swidden cultivation (see Padoch, 1982), usually within a delimited territory where forest is fallowed in a long cycle. The Iban of Wong Garai are no exception to this, using an average fallow length of about 22 years for their hill swiddens. They also cultivate rice in floodplain and swamp swiddens, where the rice varieties have longer roots and are resistant to flooding. Swamp swiddens are particularly productive per unit land and are thus fallowed for only about 2–3 years (see e.g., Dove, 1980). Given their physical environment (e.g., soils and rainfall), long fallow swid-

den cultivation is an appropriate and highly sustainable system given adequate ratios of people and land (see Cobb, 1988; Colfer *et al.*, in press).

METHOD

Data on hunting were collected by Wadley in 1993–94 using an interview schedule developed by all three authors. It was administered immediately after every hunting trip for six one-month sampling periods spread evenly over a year. Interviews were conducted in the Iban language, and animal identification was made through the use of wildlife field guides for Borneo (e.g., Payne *et al.*, 1985).⁷ Although data were collected on all aspects of hunting, the data presented here concern only adult (male) hunting of large mammals. No women participated in hunting. Hunting done by boys (generally devoted to small game) and adult hunting of birds, small mammals (such as bats and squirrels), and reptiles is not considered. Large mammals are here defined as those species whose estimated adult weight is 2.0 kg or more.

An encounter is defined as any sighting of an animal or group of animals during a hunting trip. This includes shootings and captures. Shootings refer to shooting at game without capture, while captures refer to kills in the cases here. The numbers of animals seen during encounters are usually estimates and may not accurately reflect numbers of animals in the area. A separate survey to estimate animal abundance would be desirable, such as that done by Lahm (1993a,b) in Gabon, but this was not possible given the lack of field personnel. Hunters reported (sometimes estimated) the numbers of animals seen at any one encounter, which might range from a solitary barking deer to a troop of langurs or macaques. These numbers do not represent extant population sizes because the same animals might have been encountered and therefore counted numerous times. They do, however, give some impression of the relative abundance of each species.

Forest growth in which hunting occurred was classified using indigenous categories, which correspond to important stages of forest succession: (1) *kampung*—old growth forest that has not been farmed within historical memory (this includes “virgin” forest and previously farmed forest that has returned to climax stage); (2) *pengerang tuai*—old secondary growth that

⁷Hunters were asked a series of questions regarding such things as time of day, duration of hunt, weather during hunt, location of hunt, locations traversed during hunt, type of environment in which animals were encountered, and animals encountered, shot at, and captured. Identification of captured animals brought to the longhouse was done by Wadley in order to determine species, sex, relative age, and physical condition. When there was more than one hunter on a trip, the men were interviewed separately as a cross-check; locations were identified on maps and by inspection during the course of fieldwork.

has lain fallow for 20 or more years; (3) *pengerang muda*—young secondary growth that has been fallow for 15–20 years; (4) *damun*—fallow forest about 2–15 years old; and (5) *temuda*—newly fallowed fields. Other important categories of forest include old longhouse sites (*tembawai*), old grave sites (*pendam/rarong*), and special tree reserves (*pulau*), as well as rubber tree and fruit tree groves (*kebun*).

RESULTS

Socioeconomic Data

In contrast to what Lahm (1993b) reports for Gabon and Caldecott (1988) reports for Sarawak, hunting for the market is only occasional among the Iban in this area (see also Stearman and Redford, 1992). In only one instance out of 58 successful hunts during the present study was meat sold. This was a case in which a man shot two adult bearded pigs, dividing one with the other households in the longhouse as is customary and selling the other in the market town. There is some demand for pig meat among the Christian Dayak living in Lanjak although many there also hunt themselves, and for deer meat among the Muslim Melayu if it meets certain ritual prescriptions. Primate meat is not considered worth buying or selling, but there are occasionally live orangutan and gibbon infants sold (see e.g., Meijaard *et al.*, 1996) although none were captured during the study period.

Hunting for the market is generally regarded as unrewarding given both the relatively low demand locally and the money earned (i.e., around U.S. \$1.00–1.50 per kilogram). More often men leave the area for East Malaysia (i.e., Sarawak and Sabah) or Brunei to engage in long-term wage labor, some earning more in 1 month than they could make in a year of intensive hunting if they chose to do so. This itself had implications for hunting pressures. In the study community, 22.7% of the able bodied men were gone for the duration of Wadley's fieldwork (1992–94), and during a 9-month period in 1993–94, a full 63.6% were absent on wage labor journeys. This greatly reduces the number of hunters and is a common pattern throughout the larger Iban population (see Kedit, 1993; Caldecott, 1988, p. 33).

During the study period, there were 12 adult male hunters, two of whom were particularly diligent, making over half of all hunts. An average of 1.63 hunters went out on each hunting trip, spending an average of 3.3 hours per trip. Hunting was both a diurnal and nocturnal activity with the latter being made more efficient with the use of battery powered flashlights (e.g., Caldecott, 1988). Civets, porcupine, and mousedeer were common nighttime prey.

The preferred implement for hunting large game was the 12-gauge shotgun. It has replaced the traditional spear and blowpipe, and probably reduced the use of traps as few were recorded in use during the study period. Of the 34 large mammals captured, all but one were killed with shotguns. Ammunition can be purchased at shops in the nearby market town or across the border in Sarawak; shells can also be handmade, although the product is generally unreliable. With the increased availability of factory shotgun shells and the ability to buy them, handmade shells are less common today. In the community under study, there were seven shotguns owned by five households but available through borrowing by men with no guns. Although dogs are generally ubiquitous in Iban hunting, none were used during this study (see below).

Large Mammal Encounters

Data on hunting effort over the year revealed an average 22.2 hunting trips per month, 20.2 encounters with large mammals per month, and 122.9 hours spent hunting per month.

Encounters. Encounters with primates per month were significantly less frequent than encounters with other large mammals per month (Table I). This suggests that either, despite being encountered in larger numbers (see below), primates are better able to avoid human sighting or hunters biased their reports in favor of more preferred species.

The large mammals encountered during hunting trips included seven species of primates (Table II). The other large mammals encountered include two species of barking deer, four species of civet, and two species of mousedeer. In addition there were bearded pig, sambar deer, sun bear, and thick spined porcupine. Of the primates, the pig tailed macaque was most frequently encountered followed by gibbons and white fronted leaf monkeys (Table II). The encounters with primates represented 36.4% of the total large mammal encounters. Encounters with other large mammals stood at 63.6%, while encounters with preferred game (i.e., pig, barking deer, and sambar deer) represented 44.6%.

Table I. Number of Encounters by Month, Wong Garai, 1993–94

Month ^a	1	2	3	4	5	6	Total
Number of primate encounters	11	5	13	6	4	5	44
Number of other large mammal encounters	21	13	6	16	13	8	77
Total number of encounters	32	18	19	22	17	13	121

^a1 = Apr.–May '93, 2 = June–July '93, 3 = Aug.–Sept. '93, 4 = Oct.–Nov. '93, 5 = Dec. '93–Jan. '94, 6 = Feb.–Mar. '94. Primate encounters vs. Other encounters: $\chi^2 = 11.2$, $df = 5$, $p = 0.05$.

Table II. Large Mammal Encounters, Wong Garai, 1993–94^a

		Number of Encounters (%)	Resulting in capture
A. Primates			
Gibbon	<i>Hylobates muelleri</i>	9 (20.5)	1 (14.3)
Long-tailed macaque	<i>Macaca fascicularis</i>	3 (6.8)	0 (0)
Pig-tailed macaque	<i>Macaca nemestrina</i>	19 (43.2)	3 (42.8)
Orangutan	<i>Pongo pygmaeus</i>	1 (2.3)	0 (0)
White-fronted leaf monkey	<i>Presbytis frontata</i>	8 (18.2)	2 (28.6)
Banded leaf monkey	<i>Presbytis melalophos cruciger</i>	2 (4.5)	1 (14.3)
Unidentified leaf monkey	<i>Presbytis</i> spp.	2 (4.5)	0 (0)
Total primate encounters		44 (100)	7 (100)
B. Other large mammals			
Bearcat civet	<i>Arctictis binturong</i>	1 (1.3)	1 (4.2)
Small toothed palm civet	<i>Arctogalidia trivirgata</i>	10 (12.9)	6 (25.0)
Sambar deer ^b	<i>Cervus unicolor</i>	1 (1.3)	0 (0)
Sun bear	<i>Helarctos malayanus</i>	1 (1.3)	0 (0)
Yellow barking deer ^b	<i>Muntiacus atherodes</i>	1 (1.3)	0 (0)
Red barking deer ^b	<i>Muntiacus muntjac</i>	13 (16.9)	5 (20.8)
Unidentified barking deer ^b	<i>Muntiacus</i> spp.	6 (7.8)	0 (0)
Masked palm civet	<i>Paguma larvata</i>	1 (1.3)	0 (0)
Common palm civet	<i>Paradoxurus hermaphroditus</i>	1 (1.3)	0 (0)
Bearded pig ^b	<i>Sus barbanus</i>	33 (42.9)	7 (29.1)
Thick spined porcupine	<i>Thecurus crassispinis</i>	1 (1.3)	1 (4.2)
Lesser mousedeer	<i>Tragulus javanicus</i>	1 (1.3)	1 (4.2)
Greater mousedeer	<i>Tragulus napu</i>	4 (5.2)	2 (8.3)
Unidentified mousedeer	<i>Tragulus</i> spp.	1 (1.3)	1 (4.2)
Unidentified civet		2 (2.6)	0 (0)
Total other large mammal encounters		77 (100)	24 (100)
C. Total large mammals		121	31

^aPrimates vs. other large mammals: $\chi^2 = 2.1$, $df = 1$, n.s.; primates vs. preferred game: $\chi^2 = 0.43$, $df = 1$, n.s.

^bPreferred species.

The percentage of encounters with primates resulting in capture (15.9%), while lower than that of encounters with other large mammals resulting in capture (31.2%), is not significantly different ($\chi^2 = 2.1$, $df = 1$). There is also no significant difference in this measure comparing primates and preferred game ($\chi^2 = 0.43$, $df = 1$). Thus, encounters with primates are almost as likely to result in capture as encounters with other large mammals, even though primates are less frequently encountered.

Trips. Of the 118 hunting trips taken by adult men in which large mammals were encountered, 38 (32.2%) ended in capture (Table III). This capture rate was not significantly different from that of large mammals captured (36.5%) or primates captured (21.2%). Hunting trips were thus equally likely to result in the capture of any animal, whether primates or other large mammals.

Table III. Number of Hunting Trips, Wong Garai, 1993–94^a

	Trips with encounters, No. (%)	Trips with captures, No. (%)
Primates	33 (100)	7 (21.2)
Large mammals	85 (100)	31 (36.5)
Total ^b	118 (100)	38 (32.2)

^a $\chi^2 = 2.85$, $df = 2$, n.s.^b15 other trips were made with no large mammal encounters.

Large Mammal Numbers

Primates comprised 64.2% of the total number of large mammals encountered; they formed 30% of those shot at, and 23.5% of those captured (Table IV). In comparison, Lahm records primates being 19.4% or 47 out of 242 large mammals captured in Gabon (1993a, p. 94). Caldecott reports for the upriver areas in Sarawak that primates make up only 3.2% of the total number of large mammals captured (53 out of 1649); the largest percentage (78.7) is made up of bearded pigs (1988, p. 36). Likewise, Chin gives a percentage of 5.3 (4 out of 75) for primates captured by the Kenyah of the upper Baram in Sarawak. In contrast pigs make up 65% of the total (1985, p. 103). In the Wong Garai data, the number of pigs captured was the same as that for primates, representing 23.5% of the total (8 out of 34); other preferred game (only barking deer in this case) made up only an additional 14.7% or 5 out of 34. Bodmer *et al.* (1988) also find similar capture rates for primates and ungulates in Peru.

One reason for the differences between the Wong Garai primate and preferred game capture rates and those from Sarawak may have to do with human population density as both areas have experienced both disruption from logging and the use of shotguns. The Sarawak data reported by Caldecott (1988) and Chin (1985) have human population densities of around 2–4 persons per km² with settlements fairly distant from each other. In

Table IV. Number of Large Mammals Encountered, Wong Garai, 1993–94^a

	Shot at, No. (%)	Captured, No. (%)	Total, No. (%)
Primates	9 (30)	8 (23.5)	310 (64.2)
Other	21 (70)	26 (76.5)	173 (35.8)
Total	30 (100)	34 (100)	483 (100)

^aPrimates vs. other large mammals: $\chi^2 = 33.4$, $df = 2$, $p < 0.005$.

contrast, the Wong Garai territory (with a density of 4.1 persons per km² if only the Wong Garai longhouse is counted) is occasionally hunted by members of at least five other longhouses and residents of Lanjak.⁸ The estimated density of the district (with an area of 1333 km²) in 1993 was 3.25 persons per km², while that of the vicinity of Wong Garai (with an area of around 322 km²) was much higher at 5.09 persons per km². This seems similar to what Lahm reports for Gabon although she gives no comparable figures, and it agrees with what Alvard (1995) finds for Peru.

Another reason may be the little understood animal population processes in the area. In 1995, for example, the hunters of Wong Garai reported an increase in the number of pigs encountered and killed, which suggests that the population was on an up-cycle (see Caldecott, 1991).

In the Wong Garai data, the numbers of primates shot at and captured, compared to other large mammals, were negatively related to the numbers encountered. Thus, despite being encountered in greater numbers, primates were less likely to be shot at and captured than other large mammals. (This appears to contradict the other findings about capture, but the other findings concern the number of *encounters* and *hunting trips* resulting in capture, rather than the number of actual *animals* captured.)

As primates were seen in greater numbers than other large mammals, it suggests that there was little bias in reported sightings, given that most primate species except orangutans and gibbons move in troops. Reported numbers of primates might be lower if hunters were biasing their reports in favor of more preferred game, although there is some question regarding the number of reported encounters (see above). Additionally, the fact that primates are less likely to be shot at and captured despite being encountered in greater numbers may reflect a greater difficulty in capturing them (cf. Bodmer *et al.*, 1988).

Habitat

Old growth and old secondary forest were the most important habitats for both primates and other large mammals, and most hunts and encounters occur in these forest types (Table V). The number of trips in each forest type was positively associated with the number of encounters ($K = 19$, $p = 0.008$ [Kendall's test for independence]). Hunters explicitly travel

⁸Each longhouse has its own fairly well defined territory, access to which is controlled by the community as a whole (see Wadley, 1997), and men wanting to hunt in another longhouse territory generally must seek permission from the community or accompany a community member. Outsiders without permission do not hunt within the core of another territory, but rather along its boundary. It is impractical, however, for a longhouse to patrol its territorial boundaries.

Table V. Habitat and Encounters, Wong Garai, 1993–94^a

	Number of trips, ^b No. (%)	Number of encounters, No. (%)	No. primate encounters, No. (%)	No. other encounters, No. (%)
Old growth	73 (54.8)	58 (47.9)	23 (52.3)	35 (45.5)
Old secondary	33 (24.8)	41 (33.9)	20 (45.5)	21 (27.3)
Fallow	11 (8.3)	12 (9.9)	1 (2.3)	12 (14.3)
Fruit/rubber	9 (6.8)	8 (6.6)	0 (0)	8 (10.4)
Rice field	7 (5.3)	2 (1.7)	0 (0)	2 (2.6)
Total	133 (100)	121 (100)	44 (100)	77 (100)

^aForest type/trips vs. encounters: $K = 10$, $p = 0.008$ (Kendall's test for independence, Hollander and Wolfe, 1973, pp. 185–193). Old growth vs. old secondary/primates vs. other: $\chi^2 = 0.15$, $df = 1$, n.s.

^bRepresents all trips including trips with no encounters.

to those places where game animals are likely to be found, and they also use fruiting patterns of forest trees as clues to the location of game. For instance, hunters wait to ambush game near fruiting trees, such as species of *Shorea* or *Lithocarpus* found in older growth forest.

Primates and other large mammals were equally likely to be encountered in old growth and old secondary forest ($\chi^2 = 0.82$, $df = 1$). Of the encounters with primates, 52.3% of them occurred in what Iban classified as old growth, 45.5% were in old secondary forest, and only one encounter with a primate was in an agricultural environment.⁹ The last encounter was with a group of pigtailed macaques, which are notorious among farmers for raiding crops. A similar pattern was seen with the other large mammals, where 45.5% of encounters occurred in old growth and 27.3% in old secondary forest. Pigs and civets made up the majority of encounters in agricultural zones.

Lahm (1993a,b) reports moderate to strong associations of animal species with secondary forest as well as the presence of many primate and ungulate species in relatively undisturbed old secondary forest. Likewise, Wilkie (1989) finds that where there is low population density and low incidence of market hunting, large areas of secondary forest surrounding horticulturalist settlements in Zaire are frequently and successfully hunted. He reports that capture rates are similar in secondary and old growth forest. These findings agree somewhat with the data reported here. There are some reports, however, that orangutan, gibbons, and leaf monkeys may not fare well under conditions of "agricultural mosaic" (Johns and Skorupa, 1987; Marsh *et al.*, 1987; but see Salafsky, 1993, p. 929).

⁹Given the small number of reported encounters, "agricultural zone" refers to fallow (which includes young secondary [*pengerang muda*] and young fallow [*damun and temuda*]), field (*umai*), and fruit or rubber grove (*kebun*).

The low rate of encounters in agricultural zones for both primates and other large mammals would seem to support the argument that agricultural activity disrupts game populations (see e.g., Eder, 1978, p. 68; Hart, 1978, p. 336; Estioko-Griffin, 1981, p. 63; Hames and Vickers, 1982, p. 363-364; Saffirio and Scaglione, 1982; Hill and Hawkes, 1983, p. 143; Milton, 1984, pp. 14-15; Baksh, 1985, pp. 150-151; Paolisso and Sackett, 1985, p. 186). It would also appear to go against Peterson's "ecotones" hypothesis (1981, 1982) in which she argues that the expansion of agricultural fields creates an edge effect with surrounding forest, thereby supporting a higher density of game (see Linares, 1976). Her data show one Agta community where forest clearing is "both more extensive and more intensive" and where they hunt close to their clearings, while two others that are not as involved in cultivation show an opposite pattern (1981, p. 10). The relevant distinction is the greater involvement of the one community in agriculture and the larger area of clearing in their area. Indeed, the swidden farming Palanan in Agta areas tend to hunt close to their fields (Estioko-Griffin and Griffin, 1981, p. 63), and the Ka'apor of Brazil provide some support for Peterson's hypothesis (Balée, 1994, p. 121).

Rai (1990, pp. 100-102) argues, however, that the notions of ecotones and edge effect are not strongly supported empirically, and he contradicts Peterson's claim that pig and deer are edge species, showing that Agta in the area he worked hunt these preferred species far away from agricultural sites. Vickers (1988) notes, however, that the "game depletion near settlements" hypothesis should be questioned as well: He argues that there are no long-term data on game depletion and no attention to species-level processes; there have been underestimations of game population maintenance, growth, and recruitment from outside the areas under study at any particular time. He shows that over a 10-year study period in the Amazon, variation in hunting yields of peccaries is due to the occasional movements of large herds through the area, and not from overhunting. Hunting yields of peccaries are also unrelated to human population size, and while there is a depletion of some animal species as settlements grow, there is not in others. (He is, however, focused on human populations with substantial hunting areas, i.e., 250 people or less with a hunting area of 1000 km².)

Vickers' findings suggest the possibility that all studies done to date (including this one), which rely on short-term data, may not capture accurately the interactions between people and game animals over a longer period of time. The low incidence of both primates and other large mammals encountered in agricultural zones in the Wong Garai data may be the result of these animals being able to find food in forest areas. Indeed, in the year following this study, people reported an increase in pig and monkey incursions into their fields. They attributed it to there not being enough forest

Table VI. Encounters in Old Growth Forest, Wong Garai, 1993–94^a

	Number of trips, No. (%)	Number of encounters, No. (%)	No. primate encounters, No. (%)	No. other encounters, No. (%)
Unfarmed old growth	26 (35.6)	25 (43.1)	16 (69.6)	9 (25.7)
Other old growth	47 (64.4)	33 (56.9)	7 (30.4)	26 (74.3)

^aUnfarmed old growth vs. other old growth/primates vs. other: $\chi^2 = 10.9$, $df = 1$, $p < 0.005$.

Unfarmed old growth vs. other old growth/trips vs. encounters: $\chi^2 = 0.75$, $df = 1$, n.s.

fruits for them to eat. Furthermore, the population processes of the most preferred game, pig, are not adequately understood in this area (cf. Caldecott, 1991). The period of the study might have been one in which the pig population was in a low cycle, resulting in a perceived decline in capture, increased hunting of less preferred species, and nostalgia for the “old days” (see below). As mentioned above, the hunters of Wong Garai reported an increase in the number of pigs encountered and killed in 1995, which suggests that the population was on an up-cycle. Thus, both Peterson’s ecotones hypothesis and the game depletion hypothesis must await better testing with long-term data.¹⁰

For Iban, old growth forest is not just one undifferentiated forest type. It is divided into unfarmed (climax) forest (*kampong*); old longhouse sites (*tembawai*), the oldest of which dates to the 1860s; old tree reserves (*pulau*) which are specially preserved stands of trees; and old forest cemeteries (*pendam/rarong*), some of which date back over 100 years. Here a difference between primates and other large mammals was seen (Table VI). Primates were more likely than other large mammals to be encountered in unfarmed old growth forest than in other old growth forest, although the proportions of total encounters and trips in each type of old growth are not significantly different.

Encounters with primates in unfarmed old growth forest were 69.6% of the total, while only 25.7% of encounters with other large mammals occurred in such forest. This may reflect the generally isolated nature of the other old forest growth. Although old longhouse sites, old tree reserves and old grave sites may be several hectares in size, they tend to be sur-

¹⁰There is the additional problem of different measurements used: Peterson used distance from clearings to test her hypothesis, while others cite distance from settlements which does not necessarily correspond to distance from clearings. In the Wong Garai example, the longhouse is within 1 km of old secondary and primary forest where much hunting occurs but can be as far away as 3–4 km from fields. Another problem is the settlement conditions—is the example here of a long-settled swidden-based people occupying a small home territory characterized by a mosaic of fields, fallow forest, managed forest, and older growth forest comparable to those of current or former hunter-gatherers who practice some swiddening (e.g., the Agta)?

rounded by varying stages of forest regrowth, some being less accessible to primates such as gibbon and orangutan (see Marsh *et al.*, 1987; Shafer, 1990). These sites serve as attractors to some game species such as pig, and hunters go to these sites when trees are fruiting with the explicit purpose of ambushing foraging animals (see e.g., Redford *et al.*, 1992).

Summary of Analysis

We found that although encounters with primates were significantly less frequent than encounters with other large mammals, they were as likely to result in capture as encounters with other large mammals and preferred game. Hunting trips were also equally likely to end in the capture of primates and other large mammals. This suggests a certain opportunism in hunting (see below). In contrast to these findings, however, *fewer* primates were shot at and captured in proportion to the *higher* number encountered, possibly reflecting a greater difficulty in capturing them. Indeed, the fact that primates are encountered in larger numbers might help to explain why they are as likely as other large mammals to be captured when encountered—there are more animals to shoot at. Of particular interest for forest management (see below) are the findings on habitat: Old growth and old secondary forest were the main locations of both primates and other large mammals encountered during hunts, while primates were more frequently encountered in unfarmed old growth forest.

Attitudes and Preferences

If asked to rank game animals according to their preferences, Iban will invariably put bearded pig at the top, followed by barking deer and sambar deer. All other game species are second rate. Although there is a certain degree of opportunism in hunting, it is structured by these preferences. If there is even the slightest chance that bearded pigs are in the area, hunters will pass up easy shots at other game. For example, the one (adult male) orangutan sighted during this survey was spared certain wounding if not death, because the hunter in question was afraid that an attempted shot would scare away any pigs that might be nearby. Likewise, hunters have reported passing up easy shots at barking deer for the same reason. The large body size of pigs and deer appear to be an important criterion here, which is directly related to the amount of meat and fat available in any one capture (see Redford, 1993, pp. 235-236; also Caldecott, 1988, p. 30 on the high value placed on fat pigs).

With regard to primates, Iban attitudes are mixed. Primates are generally more difficult to hunt and there is less meat available as compared

to pig or deer. Taste is also an important factor, and most primate species rank fairly low. Many people complain about primate meat tasting gamey or not having the firm texture of pig or deer meat. Fat content is also lower among primates. Because of these reasons, more Iban (particularly women) refuse to eat primate meat than refuse preferred game.

Some primate species (particularly macaque) have been long considered agricultural pests (Low, 1990, p. 81). The pigtailed macaque is notorious in this regard—pulling up cassava plants to get at the tubers, eating swidden vegetable crops, and picking patches of rice fields clean (which happened to some fields in the 1993–94 farming year). In the lowland swamp areas where other Iban longhouses rely on swamp rice cultivation, the proboscis monkey (*Nasalis larvatus*) is sometimes accused of destroying rice crops, not by picking the grain but by plucking the plant and eating the inner stalk.

The orangutan, increasingly rare in the study area, is occasionally accused of destroying some durian (*Durio* spp.) crops by throwing unripe fruit to the ground, but there is no hint of retaliation by people so affected. Meijaard *et al.* (1996, p. 2) claim that orangutans are killed as agricultural pests, but there is no evidence in any Dayak areas that they ever bother people's subsistence crops (see e.g., Caldecott, 1988, p. 40). Gibbons are at risk in the area for a number of reasons (including the shooting of mothers to capture live infants for sale), but they are considered "good" animals, destroying neither fruit nor agricultural crops. They also provide some indication of the weather in the coming day through their early morning vocalizations.

Taboos

Religious taboos against killing some species (particularly primates) by certain families may have spared them from being hunted in the past (see e.g., Mittermeier, 1987, pp. 127–128; Caldecott, 1988, p. 32). Two families in the longhouse under study claimed two different taboos against killing orangutans and gibbons, respectively. It was variously claimed that a noteworthy ancestor had been helped in battle by orangutans or gibbons, and that he had been transformed into one or the other upon death. Such taboos, however, are being set aside as people move away from their past traditions.

Conservationists have suggested the promotion of religious taboos as an avenue to preserve both animals and habitat (cf. Hames, 1991). The jury is still out on the effectiveness of indigenous religious taboos at conserving natural resources, especially when people are willing to violate or abandon taboos that conflict with current needs (see e.g., Tuan, 1968, p. 177; Hunn, 1982; Nelson, 1982, p. 225; Thomas, 1994). Most taboos are instituted, not for conservation reasons, but for social and religious goals,

and they should be expected to change with changes in society and religion. For example, members of the one Wong Garai family that had in the past a taboo against killing and eating orangutan expressed the opinion that they were no longer subject to the taboo because they were now Christian. While encouraging the maintenance of certain taboos among indigenous people may indeed serve to conserve certain species of plants and animals, the people themselves may feel more comfortable, for social and political reasons, abandoning the proscriptions. They might feel, as do some Iban, that following traditional taboos is a sign of cultural backwardness and a lack of proper cultural progress, an unfortunate national ideology that is rampant in Indonesia. Although this is not to say that indigenous taboos cannot be avenues to promote conservation (see the example below of farming ridgetops), conservationists should be aware that such national ideologies might compete with and contradict traditional ones.

Major Threats to Wildlife

Three factors stand out as potentially important human influences on some animal populations in the area: (1) the use of shotguns in hunting; (2) the increasing human population; and (3) habitat destruction.

Shotguns. Although there is no comparative evidence *in this case* to say for certain, it is likely that the use of shotguns has increased the efficiency and deadliness of hunting. Caldecott (1988, pp. 41–42) reports that two-thirds of captured animals die by shotgun in the interior of Sarawak. In separate studies on the relative efficiencies of shotguns and traditional hunting weapons (i.e., blowgun, bow, and spear) in the Amazon, Hames (1979) and Yost and Kelley (1983) conclude that although shotguns are more costly to acquire and maintain, they are more efficient in acquiring animal protein (see also Alvard, 1995). The use of shotguns may also result in more woundings with animals maimed or dying in the forest because of the increased availability of shells. Older Iban hunters recall hunting in the past when shells were limited and expensive, they were more cautious in shooting game.

In contrast to these two studies, where shotguns are used along with traditional implements, the Iban no longer use blowguns or spears in hunting, relying exclusively on the shotgun. This may have something to do with its relative cost, the Iban having access to highly productive wage labor and owning well-established smallholdings of rubber. The cost to them of buying and maintaining a shotgun, and buying shells or the materials to make them, is thus not as high as it appears to be for the Amazonian groups reported. Prior to 20 years ago, shotguns were rare in the area, and

most of the seven shotguns owned by men in the study community were acquired within the past 10–15 years (see Hames, 1987, 1991).

The lack of dogs at Wong Garai might offset any added efficiency in hunting terrestrial species, although not arboreal or nocturnal game, which may put added pressure on primate species. The Iban, as with other Dayak groups in Borneo, generally use dogs in hunting terrestrial game, especially pig and deer; dogs are not used for arboreal game or during night hunts (see e.g., Caldecott, 1988, p. 41; also Paolisso and Sackett, 1985, p. 189; Balée, 1994, p. 57). Caldecott (1988, p. 41) shows that dogs are very important in hunting pig and deer (see e.g., Yost and Kelley, 1983, p. 206, on the importance of dogs in spear-hunting).

That being said, it may be significant that about 5 years prior to this research, all of the good hunting dogs at Wong Garai died in a distemper epidemic, and given the high value of dogs from good hunting lines, they have been unable to acquire puppies from kin in other longhouses. They complain about their dogs now not knowing how to cooperate with hunters, and of being more likely to bay at and catch a domestic pig than a forest pig. This lack of hunting dogs may result in an increase in effort (or success?) at arboreal hunting (a threat to most primates), a subsequent decline in success at terrestrial hunting, and increased opportunism.

Human Population. Alvard (1995) notes that human population increases may be more significant for the decline in game species than the use of more efficient hunting technology. It is generally recognized by the local people that their numbers are increasing, although presently at low rates. Population data from the regional and district governments (Kabupaten Kapuas Hulu, 1984; Kecamatan Batang Lupar, 1988, 1995) show that the district population grew from 4091 in 1984 to 4438 in 1995 (an annual rate of 0.72%). Within the vicinity of Wong Garai (which includes nine longhouses and the town of Lanjak), the population grew from 1589 in 1988 to 1675 in 1995 (an annual rate of 0.73%). The increase is especially seen in Lanjak, to which people are moving from outside and from within the district to take advantage of the new government road and schools. Such increases in population may give rise to more hunting for the market (see Wilkie, 1989), but only if the added population is not inclined to hunting.

In the longhouse under study, the rate of growth has been fairly slow with the population nearly doubling in 15–20 years, and the number of households increasing from 7 to 14 in 15 years. While this increase in population may have also increased the demand on forest resources, and farming may have expanded the area of forest land under the cultivation, these speculations remain to be demonstrated. Additionally, the 14 hunters from this one longhouse were not the only ones hunting in the area—a number of Iban and other Dayak from the market town and men from at least five

other longhouses nearby also made use of the forests that lie within their own and the study longhouse's territories.

Within the community of Wong Garai, the population may stabilize or even decline in the future, despite its past growth. For about a decade, the reproductively active women have taken very readily to the increased availability of government family planning programs, particularly the use of contraception. The women view the ability to control their pregnancies as very positive because it not only reduces the very real risk of death during labor, but it also relieves them of the increased burden of childcare and allows them to be more economically productive. With women thus having fewer children (and some of those children, having been through state school, finding employment away from the area), the local human population may well decline. That might be good news for primate and other animal populations.

The threat of increased migration into the area, however, may counteract any decline in native population. Under Indonesia's controversial program of transmigration whereby people from the populous islands of Java and Bali are resettled in less populated islands like Kalimantan, the district in which Wong Garai is located is expected to receive some 200 families within the next several years. They are to be settled in the middle Empasuk floodplain (some 6 km from Wong Garai) and provided with land allotments for rice farming. These allotments might cut into existing (but previously logged) lowland forest, threatening both habitat for animals and hunting areas for locals. Although there should be no increased demand for market hunting (given that most transmigrants will probably be Muslim), the resulting disruption of forest habitat through conversion to agriculture may be highly detrimental to animal populations. A case from East Kalimantan is instructive: After transmigrants settled in a forest concession in which the Uma' Jalan Kenyah community of Long Segar was located, the population density rose from 2–4 persons per km² to 60 persons per km² (Colfer, 1995).

Habitat Destruction. Most studies to date have concentrated on the effects of so-called selective logging on animal populations in tropical forests (e.g., Wilson and Wilson, 1975; Wilson and Johns, 1982; Johns, 1985; Skorupa, 1986; Marsh *et al.*, 1987). The commercial harvesting of timber in the study area, however, has an unknown effect on the local fauna, but it is likely to be largely negative given the intensity of the cutting (see Johns, 1983, 1988; Cannon *et al.*, 1994). Censuses show, for example, that orangutan numbers are unusually high in the reserve area, the possible result of animals fleeing disruption from logging activities around the reserve (Meijaard *et al.*, 1996; Russon *et al.*, 1996).

One indication of logging's effects on animal populations may be seen in returns to hunting before and after logging. Caldecott reports a sharp

decline in wildlife harvests within 10 years of an area being logged with a new low equilibria of some species and further decline of others over the next two decades. This is the result of habitat destruction, increased demands for wild meat by logging camps, better access to forests from logging roads, and greater availability of ammunition and lights (1988, pp. 90-91).

People's experience in Long Segar, East Kalimantan, may be a harbinger of bad news for Iban hunters and local game populations. There, logging began in the mid-1970s, complicating the past land use history of natural forests and swidden cultivation; this was followed by the introduction of industrial timber plantations in 1992. Uma' Jalan Dayaks complained to Colfer in 1995 that there were far fewer pigs than there used to be, and that those captured had almost no fat (see above). Indeed, their traditional system was undergoing a "frontal attack" from competing land uses, like mining, logging, plantations, and transmigration.

Blame is often placed on swidden cultivation for much forest destruction (see e.g., Wilson and Wilson, 1975; MacKinnon, 1986; Marsh *et al.*, 1987; Meyers, 1987). While this is a view held by governments and some conservationists, it stems from a lack of knowledge or appreciation for the actual complexity of indigenous agroforestry practices (see Clay, 1988; Alcorn, 1990; Redford and Padoch, 1992; Colfer and Dudley, 1993; Colfer *et al.*, in press; Brookfield and Padoch, 1994). In addition to forest fallow at different stages of regrowth (see the classifications above), there are important sites of managed forest resources (e.g., old longhouse sites, see Sather, 1990). These practices have advantages over many competing land uses in that soils are better protected, local watersheds are maintained, and habitat is available to some fauna (Padoch and Peters, 1993). With a mix of unfarmed old growth forest, various stages of forest regrowth after farming, and forest gardens, the total floral and faunal diversity may well be higher compared to "pristine" old growth (see Balée, 1994, pp. 135-138). This is a topic that urgently needs more study.

CONSERVATION AND WILDLIFE MANAGEMENT

Ultimately, effective wildlife management and conservation will have to be supported by local people. The legitimacy and importance of their concerns and needs are increasingly recognized in international settings (see Caldecott, 1988; Padoch and Peters, 1993; Rainforest Alliance, 1993; Forest Stewardship Council, 1994; Greenpeace, 1994; Soil Association, 1994; Colfer *et al.*, 1995). Pragmatically, in the Kalimantan setting, we have difficulty imagining how wildlife management in particular, and conservation in general, can occur without their involvement and commitment. There already exists a body of national law regulating hunting and requiring various conservation

measures; there is an agency with a mandate and personnel responsible for ensuring forest protection and nature conservation; there are police and governmental agencies with the authority (on paper) to enforce national and regional laws and regulations; and there is even an externally funded conservation project in operation in the area. Yet local community resource management is, in fact, the only management in operation.

That the Iban actively manage their forest resources is not in doubt (see e.g., Padoch, 1982; Cramb, 1989; Sather, 1990; Wadley, 1997). Indeed, they are part of the growing list of indigenous cultivators around the world who are known to do so (e.g., Posey, 1985; Colfer and Dudley, 1993; Colfer *et al.*, in press). In contrast to what Balée (1994, pp. 138-141) claims for the Ka'apor of Brazil, the Iban are explicitly aware that their actions and practices have consequences for their descendants, and they actively pursue strategies of land use (e.g., prohibiting logging or farming in certain areas of forest, planting fruit trees, etc.) with the future in mind.

The Iban themselves are aware of the decline (or at least the potential for decline) in game species and speak glowingly of the days when they would set off for the fields in the morning and end up capturing one or two pigs every other day. While some of this may be exaggerated nostalgia (and some may be lack of awareness of long term cycles in pig populations), it is reasonable to conclude that hunting was better in the past given the potential for shotgun use, logging, and human population growth to disrupt animal population and habitats. Some hunters have expressed the need to preserve more forest from both farming and lumbering in order to maintain important hunting resources. Indeed, Wong Garai has taken a very unusual step in this direction, entirely on its own. They have set aside a large tract of forest, making it unavailable for farming. They have managed to negotiate a prohibition on logging within this area with the timber concession in which their community falls. Their purpose was to preserve a fresh water supply, a scenic waterfall for tourists, and game animal populations. Other animals and plants will, of course, also benefit from such protection.¹¹

The Iban of Wong Garai recognize as a problem the potential for increasing game scarcity and its effect on populations of less desirable species. That they see the links among hunting, logging, wildlife, and habitat destruction is clear from the particular action they have taken to address the problem. The fact that they have successfully regulated their own community land use, and negotiated an agreement with the concessionaire, to their own (and conservationists') advantage represents evidence of impor-

¹¹Wong Garai was fortunate to have one of its members (who was also a partner in a different local timber concession) in the regency legislature at the time. It was with his help and influence on regency and district governments that Wong Garai was successful.

tant "social capital" (Ostrom, 1994) that can and should be recognized and used in formal conservation efforts. Such capital represents an opportunity for those, such as the Danau Sentarum Wild Reserve conservation project and the Agency for Forest Protection and Nature Conservation (PHPA), who manage the area with wildlife and other conservation considerations in mind (see also Colfer *et al.*, 1996).

To harness such human capital in improving wildlife management and conservation in the area, it is necessary to assess the motivations and values behind local hunting and land use (see Palmer, 1993). The primary aim of Iban hunting and land use is to procure food and materials for their families. If improved resource management (such as more controlled hunting or larger protected tracts of forest) will result in improved access to food and other benefits to the Iban, they are likely to be interested.

Although there is a difference in priorities between most conservationists and the Iban, with the former focused on the environment and the latter on their livelihoods, the two groups share a number of important concerns. From the standpoints of human well-being among the Iban and wildlife conservation, hunting issues cannot be separated from land use. Preservation of remaining old forests and promotion of land use practices that lead to more protected old growth and more long fallowed secondary forest, shown to be important in these data, should benefit both conservation and the Iban. The Iban, like most conservationists, are aware of the potential for increasing land shortages as the population increases. Both are also increasingly aware of the level of profits available to the relatively unregulated timber concessionaires in the area. The fact that the Iban have a strong sense of place and concern for their traditional land rights adds another dimension to their awareness. Iban land use systems (i.e., based on swidden cultivation), under continual fire from the government as "primitive" and "destructive," are in fact part of a complex agroforestry system which contains many opportunities for supporting conservation goals—something many conservationists do not realize.

There are a variety of general actions that seem intuitively to contribute to habitat viability in this context:

- enforcement of existing forestry regulations on logging in the area, including reductions in allowable logging and monitoring of compliance with regulations,
- enhancement of existing programs to encourage birth control, and the cessation of government-sponsored and large-scale spontaneous in-migration,
- reduction of hill swidden size (and thus an increase in old secondary forest size) through increasing and intensifying swamp

- rice farming¹² and expansion of vegetable gardening beyond hill swiddens where vegetables are traditionally grown,
- enhancement of existing managed orchards and forest areas, including supplementary fruit planting for animals (see Russon *et al.*, 1996),
 - preservation of plant species not normally used by people, for the purposes of maintaining biodiversity and attracting games and other species (see Redford *et al.*, 1992; cf. Lawrence *et al.*, 1995),
 - and promotion of programs to increase domestic meat production for both home consumption and sale.¹³

From the standpoint of wildlife management as well, there are useful actions that could enhance the availability of wildlife for hunting and serve to protect endangered species. Typical mechanisms consistent with conventional, “modern” wildlife management approaches include regulation of firearms (something to which the Iban have been highly resistant in the past), restriction of ammunition supply (see Caldecott, 1988, p. 110), restriction on seasons, and restrictions on amount, age and/or sex of allowable catch (see e.g., Bodmer *et al.*, 1988). The main problem with these approaches as conventionally applied is that they tend to increase the complexity of government bureaucracy, thus removing control from locals (see e.g., Tainter, 1996).

Together, an increase in protected forest habitat and a decrease in hunting efficiency might aid in the growth of the preferred game populations, particularly pig (see Feer, 1993). We believe this would take pressure off secondary game species, such as primates. It is probable that the principle reason that Iban are hunting other species is not a desire to do so, but rather because of the decline in preferred game populations. Where more valuable game is not easily available, less valued game will be sought in order to provide occasional animal protein to the diet (see Stearman, 1992; Vickers, 1988). Colfer *et al.* (in press) have direct dietary evidence from four Kenyah villages, suggesting this kind of process is occurring in East Kalimantan as well. Iban themselves maintain that they would have rarely bothered hunting most animals other than pigs in the past. Addi-

¹²This may occur anyway with an irrigation dam and canals currently under construction within Wong Garai territory; it is aimed at promoting irrigated rice cultivation among the local Dayak and providing the eventual transmigrants with irrigation water.

¹³There is currently no government support for improving domestic pig production, given the national and regional governments' domination by Muslims. There are, however, programs for cattle, goats, and chicken husbandry, and Wong Garai has received a number of cows in the past with generally poor results. The area lacks appropriate fodder, and the people are constrained by their subsistence labor to provision cattle. The most appropriate animals, given the environment and the people's cultural preferences, are those they already raise—pigs and chickens.

tional studies of game populations in the area are needed to adequately assess the real impact of hunting, particularly over the long term (Robinson and Redford, 1994; Racz-Luna, 1995).

The creation of forest corridors¹⁴ may be necessary to connect the increasingly isolated patchwork of old growth forest in the area (see Shafer, 1990). This patchwork does not greatly affect highly mobile species such as pig or primate species like macaque, which often descend to the forest floor during foraging. It does, however, particularly affect gibbons and orangutans, resulting in isolated and limited breeding groups. Meijaard *et al.* (1996) and Russon *et al.* (1966) have recommended expanding the present reserve boundary to include a corridor linking the reserve with the Bentuang Karimun Nature Reserve situated approximately 50 km to the north; this would connect populations of orangutan and other older growth-dependent animals. The new and expanding road system in the area, especially to the north of the reserve, is an immediate threat in this regard (see Wadley, n.d.). The main government road cuts through the area between the reserves, and there are plans to build roads to the east of the Danau Sentarum reserve, which would directly threaten the implementation of forest corridors between the two reserves. One possible avenue on a small scale would be to strengthen the traditional Iban proscription (which has already received official support from the regency government) against farming along the ridge tops of mountains extending it to a proscription on farming along ridge lines as well thus linking wider areas.

All of the above are, however, simply ideas, words on paper, without substantive recognition of the involvement of local people. The only significant management currently taking place is in local people's hands. They have the greatest motivation and opportunity to manage well, and they possess a wealth of useful knowledge about local wildlife and habitats. The critical prerequisite for putting such ideas into practice is the empowerment of local people, who, *in consultation with* other stakeholders (e.g., the conservation project), can manage these resources themselves (see Colfer *et al.*, 1996). The most useful role for outsiders, like the governments and projects, is to support and strengthen existing management, helping local people to keep in mind the long-term goal of sustainable management of wildlife and other natural resources in their care.

CONCLUSIONS

We have examined the broad patterns of Iban hunting comparing non-human primates and other large mammals, finding that: (1) encounters with primates were significantly less frequent than with other large mammals;

¹⁴Forest corridors are links of forest growth between larger areas of forest, providing a way for animals limited to forest to pass from one area to another.

(2) encounters with primates were as likely to result in capture as encounters with other large mammals; (3) hunting trips were equally likely to result in capture of primates and other large mammals; (4) despite being encountered in greater numbers, fewer primates were shot at and captured than other large mammals; and (5) primates and other large mammals were equally likely to be encountered in both old growth and old secondary forest, although primates were more likely to be found in unfarmed old growth than other large mammals. This last finding is probably the most important from a management standpoint, in that it suggests an aim of local land use management in order to enhance hunting.

The finding that encounters with primates were as likely to result in capture as encounters with other large mammals, even though primates were encountered less frequently, suggests that hunters were more opportunistic than we expected based on their stated preferences. In addition, they were more opportunistic than other hunters in Borneo (see Chin, 1985; Caldecott, 1988). Their opportunism may be the result of declining numbers of preferred game, although we have only anecdotal evidence on hunting in the past. This does appear plausible in light of data reported by Caldecott (1988) and Chin (1985) on greater numbers of preferred game captured in other Borneo populations.

We have also pointed out the potential threats from shotgun use, human population increases, and forest destruction to local animal and plant populations, with some thoughts on possible ways to support sustainable local management and conservation concerns. The local people believe that they have nationally recognized rights in the land they have occupied and managed for generations (a view which is contested by some powerful stakeholders such as timber concessionaires). Although the strength of this right, in actuality, remains in question (from an external point of view), their belief in it potentially gives them an important stake in the development of conservation policies and management of local resources. The best outcome—indeed, the only one we think will work—is a situation where the local people police, monitor, and regulate themselves, taking into account the need for long-term forest maintenance (see Colfer *et al.*, 1996, in press; also Ostrom, 1990, 1994). Active involvement of local people and attention to their needs and concerns, and the willingness of all sides involved to compromise for long-term benefits, are essentials to the success of this and any other conservation effort.

ACKNOWLEDGMENTS

The authors served as consultants for the Danau Sentarum Wildlife Conservation Project under the auspices of Asian Wetland Bureau (now Wetlands International-Indonesia Programme), the Indonesian Directorate of

Forest Protection and Nature Conservation, and Overseas Development Administration (U.K.). In addition, Wadley's field research (1992–1994) was funded by the National Science Foundation (Grant No. BNS-9114652), Wenner-Gren Foundation for Anthropological Research, Sigma Xi, and Arizona State University and was sponsored by the Balai Kajian Sejarah dan Nilai Tradisional Pontianak with permits from the Lembaga Ilmu Pengetahuan Indonesia. Additional study was done by Colfer and Wadley in June–July 1995 under the auspices of the Center for International Forestry Research. We wish to thank John Alcock, James Eder, Leanne Nash, Oona Paredes, and three anonymous reviewers for their comments on earlier drafts.

REFERENCES

- Alcorn, J. B. (1990). Indigenous agroforestry systems in the Latin American tropics. In Altieri, M. A. and Hecht, S. B. (eds.), *Agroecology and small farm development*. CRC Press, Boston.
- Alvard, M. (1995). Shotguns and sustainable hunting in the Neotropics. *Oryx* 29: 58–66.
- Baksh, M. (1985). Faunal food as a "limiting factor" on Amazonian cultural behavior: A Machiguenga example. *Research in Economic Anthropology* 7: 145–175.
- Balée, W. (1994). *Footprints in the forest: Ka'apor ethnobotany—The historical ecology of plant utilization by an Amazonian people*. Columbia University Press, New York.
- Bodmer, R. E., Fang, T. G., and Ibañez, L. M. (1988). Primates and ungulates: A comparison of susceptibility to hunting. *Primate Conservation* 9: 79–83.
- Brookfield, H., and Padoch, C. (1994). Appreciating agrobiodiversity: A look at the dynamism and diversity of indigenous farming practices. *Environment* 36(5): 6–11, 37–43.
- Caldecott, J. (1988). *Hunting and wildlife management in Sarawak*. IUCN Gland, Switzerland.
- Caldecott, J. (1991). Eruptions and migrations of bearded pig populations. *Bongo* 18: 233–243.
- Cannon, C. H., Peart, D. R., Leighton, M., and Kartawinata, K. (1994). The structure of lowland rainforest after selective logging in West Kalimantan, Indonesia. *Forest Ecology and Management* 67: 49–68.
- Chin, S. C. (1985). Agriculture and resource utilization in a lowland rainforest Kenyah community. Special Monograph No. 4. *Sarawak Museum Journal* 35(56).
- Clay, J. (1988). *Indigenous people and tropical forests*. Cultural Survival, Cambridge, MA.
- Cobb, D. D. (1988). Iban Shifting Cultivation: A Bioregional Perspective, Ph.D. thesis, Arizona State University.
- Colfer, C. J. P. (1995). Who Counts Most in Sustainable Forest Management? Working Paper No. 7, Center for International Forestry Research, Bogor, Indonesia.
- Colfer, C. J. P., and Dudley, R. G. (1993). *Shifting cultivators of Indonesia: Marauders or managers of the forest?* FAO, Rome.
- Colfer, C. J. P., Peluso, N., and Chin S. C. *Beyond slash and burn: Lessons from the Kenyah on management of Borneo's rain forests*, New York Botanical Gardens, New York (in press).
- Colfer, C. J. P., Prabhu, R., and Wollenberg, E. (1995). Principles, Criteria and Indicators: Applying Ockham's Razor to the People-Forestry Link. Working Paper No. 8, Center for International Forestry Research, Bogor, Indonesia.
- Colfer, C. J. P., Wadley, R. L., and Widjanti, E. (1996). Using indigenous organizations from West Kalimantan. In Blunt, P., and Warren, D. M. (eds.), *Indigenous organizations and development*. Intermediate Technology, London.
- Conklin, H. C. (1957). *Hanunó agriculture: A report on an integral system of shifting cultivation in the Philippines*. FAO, Rome.

- Cramb, R. A. (1989). Explaining variations in Bornean land tenure: The Iban case. *Ethnology* 28: 277-300.
- Deschamps, V. (1995). Community-based resource management in Danau Sentarum Wildlife Reserve. *Conservation Indonesia* 11: 30-33.
- Dove, M. R. (1980). The Swamp rice swiddens of the Kantu' of West Kalimantan, Indonesia. In Furtado, J. I. (ed.), *Tropical ecology and development*. International Society of Tropical Ecology, Kuala Lumpur.
- Dove, M. R. (1981). Subsistence Strategies in Rain Forest Swidden Agriculture: The Kantu' at Tikal Batu. Ph.D. thesis, Stanford University.
- Eder, J. F. (1978). The caloric returns to food collecting: Disruption and change among the Batak of the Philippine tropical forest. *Human Ecology* 6: 55-69.
- Estioko-Griffin, A. A., and Griffin, P. B. (1981). The beginning of cultivation among Agta hunter-gatherers in northeast Luzon. In Olofson, H. (ed.), *Adaptive strategies and change in Philippine swidden-based societies*, Forest Research Institute, Laguna, Philippines.
- Feer, F. (1993). The potential for sustainable hunting and rearing of game in tropical forests. In Hladik, C. M., et al. (eds.), *Tropical forests, people and foods: Biocultural interactions and applications to development*. UNESCO and parthenon Publishing, Paris.
- Forest Stewardship Council (1994). *Forest stewardship council ratification documents (July)*. Oaxaca, Mexico.
- Freeman, D. (1970). *Report on the Iban*, Athlone Press, London.
- Giesen, W. (1987). *Danau Sentarum wildlife reserve: Inventory, ecology and management guidelines*, WWF and PHPA, Bogor.
- Greenpeace (1994). Principles and Guidelines for Ecologically Responsible Forest Use (March).
- Grenand, P. (1993). Fruits, animals and people: Hunting and fishing strategies of the Wayapi of Amazonia. In Hladik, C. M., et al. (eds.), *Tropical forests people and foods: Biocultural interactions and applications to development*. UNESCO and Parthenon Publishing, Paris.
- Hames, R. B. (1979). A comparison of the efficiencies of the shotgun and the bow in neotropical forest hunting. *Human Ecology* 7: 219-252.
- Hames, R. (1987). Game conservation or efficient hunting? In McCay, B., and Acheson, J. (eds.), *The question of the commons: The culture and ecology of communal resources*. University of Arizona Press, Tucson.
- Hames, R. (1991). Wildlife conservation in tribal societies. In Oldfield, M. L., and Alcorn, J. B. (eds.), *Biodiversity: Culture, conservation, and ecodevelopment*. Westview Press, Boulder.
- Hames, R. B., and Vickers, W. T. (1982). Optimal diet breadth theory as a model to explain variability in Amazonian hunting. *American Ethnologist* 9: 358-378.
- Hart, J. A. (1978). From subsistence to market: A case study of the Mbuti net hunters. *Human Ecology* 6: 325-353.
- Hill, K., and Hawkes, K. (1983). Neotropical hunting among the Aché of eastern Paraguay. In Hames, R. B., and Vickers, W. T. (eds.), *Adaptive responses of native amazonians*. Academic press, New York.
- Hollander, M., and Wolfe, D. A. (1973). *Nonparametric statistical methods*. John Wiley and Sons, New York.
- Hunn, E. (1982). Mobility as a factor limiting resource use in the Columbia River Plateau of North America. In Williams, N., and Hunn, E. (Eds.), *Resource managers: North American and Australian hunter-gatherers*. American Association for the Advancement of Science, Washington, D.C.
- Johns, A. D. (1983). Tropical forest primates and logging—Can they co-exist? *Oryx* 17: 114-118.
- Johns, A. D. (1985). Selective logging and wildlife conservation in tropical rain-forest: Problems and recommendations. *Biological Conservation* 31: 355-375.
- Johns, A. D. (1988). Effects of "selective" timber extraction on rain forest structure and composition and some consequences for frugivorous and folivores. *Biotropica* 20: 31-37.
- Johns, A. D., and Skorupa, J. P. (1987). Responses of rain-forest primates to habitat disturbance: A review. *International Journal of Primatology* 8: 157-191.
- Jorgenson, J. P. (1995). Maya subsistence hunters in Quintana Roo, Mexico. *Oryx* 29: 49-57.

- Kabupaten Kapuas Hulu (1984). *Kabupaten Kapuas Hulu Dalam Angka, 1984*. Kantor Statistic, Kabupaten Kapuas Hulu Daerah Tingkat II, Putussibau.
- Kecamatan Batang Lupar (1988). *Monografi Kecamatan Batang Lupar, 1988*. Kantor Camat, Kecamatan Batang Lupar, Lanjak.
- Kecamatan Batang Lupar (1995). *Data Penduduk Wilayah Kec. Batang Lupar, Oktober 1995*. Kantor Camat, Kecamatan Batang Lupar, Lanjak.
- Kedit, P. M. (1993). *Iban Bejalai*, Sarawak Literary Society and Ampang Press, Kuala Lumpur.
- King, V. T. (1978). Introduction. In King, V. T. (ed.), *Essays on Borneo societies*. Oxford University Press.
- Lahm, S. A. (1993a). Ecology and Economics of Human/Wildlife Interaction in Northeastern Gabon. Ph.D. thesis, New York University.
- Lahm, S. A. (1993b). Utilization of forest resources and local variation of wildlife populations in northeastern Gabon. In Hladik, C. M., et al. (eds.), *Tropical forests, people and foods: Biocultural interactions and applications to development*. UNESCO and Parthenon Publishing, Paris.
- Lawrence, D. C., Leighton, M., and Peart, D. R. (1995). Availability and extraction of forest products in managed and primary forest around a Dayak village in West Kalimantan, Indonesia. *Conservation Biology* 9: 76-88.
- Lehman, S. M., Prince, W., and Taylor, L. L. (1995). Habitat Disturbance, Hunting Pressures and Primate Distribution in N.E. Guyana. Paper presented at the Eighteenth Annual Meeting of the American Society of Primatologists, June 21-24, Scottsdale, Arizona.
- Linares, O. F. (1976). "Garden hunting" in the American tropics. *Human Ecology* 4: 331-349.
- Low, H. (1990). *Sarawak, its inhabitants and productions* (reprint of 1848 Ed.). Pustaka Delta Pelajaran, Petaling Jaya.
- MacKinnon, K. (1986). The conservation status of nonhuman primates in Indonesia. In Benirschke, K. (ed.), *Primates: The road to self-sustaining populations*. Springer-Verlag, New York.
- Marsh, C. W., Johns, A. D., and Ayres, J. M. (1987). Effects of habitat disturbance on rain forest primates. In Marsh, C. W., and Mittermeier, R. A. (eds.), *Primate conservation in the tropical rain forest*. Alan R. Liss, New York.
- Marsh, C. W., and Mittermeier, R. A. (1987). Introduction. In Marsh, C. W., and Mittermeier, R. A. (eds.), *Primate conservation in the tropical rain forest*. Alan R. Liss, New York, pp. xi-xviii.
- Meijaard, E., Dennis, R., and Erman, A. (1996). Orangutan Conservation in and around the Danau Sentarum Wildlife Reserve, West Kalimantan: Management Implications. The International MOF-Tropenbos Kalimantan Project and Indonesia-UK Tropical Forest Management Programme, Jakarta.
- Meyers, N. (1987). Trends in the destruction of rain forests. In Marsh, C. W., and Mittermeier, R. A. (eds.), *Primate conservation in the tropical rain forest*. Alan R. Liss, New York.
- Milton, K. (1984). Protein and carbohydrate resources of the Maku Indians of northwestern Amazonia. *American Anthropologist* 86: 7-27.
- Mittermeier, R. A. (1987). Effects of hunting on rain forest primates. In Marsh, C. W., and Mittermeier, R. A. (eds.), *Primate conservation in the tropical rain forest*. Alan R. Liss, New York.
- Nelson, R. (1982). A conservation ethic and environment: The Koyukon of Alaska. In Williams, N., and Hunn, E. (Eds.) *Resource managers: North American and Australian hunter-gatherers*. American Association for the Advancement of Science, Washington, D.C.
- Ostrom, E. (1990). *Governing the commons: The evolution of institutions for collective action*. Cambridge University Press, New York.
- Ostrom, E. (1994). Neither Market nor State: Governance of Common-Pool Resources in the Twenty-First Century. Published lecture, presented 2 June, International Food Policy Research Institute, Washington, D.C.
- Padoch, C. (1982). *Migration and its alternatives among the Iban of Sarawak*, Martinus Nijhoff, The Hague.

- Padoch, C., and Peters, C. (1993). Managed forest gardens in West Kalimantan, Indonesia. In Potter, C. S., Cohen, J. I., and Janczewski, D. (eds.) *Perspectives on biodiversity: Case studies of genetic resource conservation and development*. AAAS Press, Washington, D.C.
- Palmer, C. T. (1993). Folk management, "soft evolutionism," and fishers' motives: Implications for the regulation of the lobster fisheries of Maine and Newfoundland. *Human Organization* 52: 414-420.
- Paolisso, M., and Sackett, R. (1985). Traditional meat procurement strategies among the Irapa-Yukpa of the Venezuela-Colombia border area. *Research in Economic Anthropology* 7: 177-199.
- Payne, J., Francis, C. M., and Phillips, K. (1985). *A field guide to the mammals of Borneo*. The Sabah Society and WWF Malaysia, Kuala Lumpur.
- Peterson, J. T. (1981). Game, farming, and interethnic relations in northeastern Luzon, Philippines. *Human Ecology*, 9: 1-22.
- Peterson, J. T. (1982). The effects of farming expansion on hunting. *Philippine Sociological Review* 30: 35-50.
- Posey, D. (1985). Indigenous management of tropical forest ecosystems: The case of the Kayapó Indians of the Brazilian Amazon. *Agroforestry Systems* 3: 139-158.
- Raez-Luna, E. F. (1995). Hunting large primates and conservation of the Neotropical rain forests. *Oryx* 29: 43-48.
- Rai, N. K. (1990). Living in a Lean-to: Philippine Negrito Foragers in Transition. Anthropological Papers No. 80, Department of Anthropology, University of Michigan, Ann Arbor.
- Rainforest Alliance (1993). *Smart wood certification program: Generic guidelines for assisting natural forest management* (revised draft, October), New York.
- Redford, K. H. (1993). Hunting in neotropical forests: A subsidy from nature. In Hladik, C. M., et al. (eds.), *Tropical forests, people and foods: Biocultural interactions and applications to development*. UNESCO and Parthenon Publishing, Paris.
- Redford, K. H., and Padoch, C. (eds.) (1992). *Conservation of neotropical forests: Working from traditional resource use*. Columbia University Press, New York.
- Redford, K. H., Klein, B., and Murcia, C. (1992). Incorporation of game animals into small-scale agroforestry systems in the neotropics. In Redford, K. H., and Padoch, C. (eds.), *Conservation of neotropical forests: Working from traditional resource use*. Columbia University Press, New York.
- Robinson, J. G., and Redford, K. H. (1994). Measuring the sustainability of hunting in tropical forests. *Oryx* 28: 249-256.
- Russon, A. E., Erman, A., and Sinaga, P. (1996). *Nest-count survey of Orangutans (Pongo pygmaeus pygmaeus) in and around the Danau Sentarum Wetlands Reserve, West Kalimantan*. Wetlands International, Pontianak, Indonesia.
- Saffirio, G., and Scaglione, R. (1982). Hunting efficiency in acculturated and unacculturated Yanomama villages. *Journal of Anthropological Research* 38: 315-327.
- Salafsky, N. (1993). Mammalian use of a buffer zone agroforestry system bordering Gunung Palung National Park, West Kalimantan, Indonesia. *Conservation Biology* 7: 928-933.
- Sather, C. (1990). Trees and tree tenure in Paku Iban society: The management of secondary forest resources in long-established Iban community. *Borneo Review* 1: 16-40.
- Shafer, C. L. (1990). *Nature reserves: Island theory and conservation practice*. Smithsonian Institution, Washington and London.
- Skorupa, J. (1986). Responses of rain forest primates to selective logging in Kibale Forest, Uganda: A summary report. In Benirschke, K. (ed.), *Primates: The road to self-sustaining populations*. Springer-Verlag, New York.
- Soil Association (1994). *Responsible forestry standards (February)*. U.K.
- Stearman, A. M. (1992). Neotropical indigenous hunters and their neighbors: Siriono, Chiriguano, and Yuqui hunting on the Bolivian frontier. In Redford, K. H., and Padoch, C. (eds.), *Conservation of neotropical forests: Working from traditional resource use*. Columbia University Press, New York.
- Stearman, A. M., and Redford, K. H. (1992). Commercial hunting by subsistence hunters: Siriono Indians and Paraguayan caiman in lowland Bolivia. *Human Organization* 51: 235-244.

- Tainter, J. A. (1996). Cultural Conflict and Sustainable Development: Managing Subsistence Hunting in Alaska. Paper presented at the Annual Conference of the International Boreal Forest Research Association, St. Petersburg, 19-23 August.
- Takeda, J., and Sato, H. (1993). Multiple subsistence strategies and protein resources of horticulturalists in the Zaire basin: The Ngandu and the Boyela. In Hladik, C. M., *et al.* (eds.), *Tropical forests people and foods: Biocultural interactions and applications to development*. UNESCO and parthenon Publishing, Paris.
- Thomas, W. H. (1994). Are Indigenous People Ecologically Wise? (unpublished ms.). Department of Anthropology, Arizona State University.
- Tuan, Y. (1968). Discrepancies between environmental attitude and behavior: Examples from Europe and China. *Canadian Geographer* 12: 176-191.
- Vickers, W. T. (1988). Game depletion hypothesis of Amazonian adaptation: Data from a native community. *Science* 239: 1521-1522.
- Vickers, W. T. (1993). Changing tropical forest resource management strategies among the Siona and Secoya Indians. In Hladik, C. M., *et al.* (eds.), *Tropical forests, people and foods: Biocultural interactions and applications to development*. UNESCO and Parthenon Publishing, Paris.
- Wadley, R. L. (1997). Variation and changing tradition in Iban land tenure. *Borneo Research Bulletin* 28 (in press).
- Wadley, R. L. (n.d.). The Road to Change in the Kapuas Hulu Borderlands: Jalan Lintas Utara (unpublished ms.).
- Wilkie, D. S. (1989). Impact of roadside agriculture on subsistence hunting in the Ituri forest of northeastern Zaire. *American Journal of Physical Anthropology* 78: 485-494.
- Wilson, C. C., and Wilson, W. L. (1975). The influence of selective logging on primates and some other animals in East Kalimantan. *Folia Primatologica* 23: 245-274.
- Wilson, W. L., and Johns, A. D. (1982). Diversity and abundance of selected animal species in undisturbed forest, selectively logged forest and plantations in East Kalimantan, Indonesia. *Biological Conservation* 24: 205-218.
- Yost, J. A., and Kelley, P. M. (1983). Shotguns, blowguns, and spears: The analysis of technological efficiency. In Hames, R. B., and Vickers, W. T. (eds.), *Adaptive responses of native Amazonians*. Academic Press, New York.